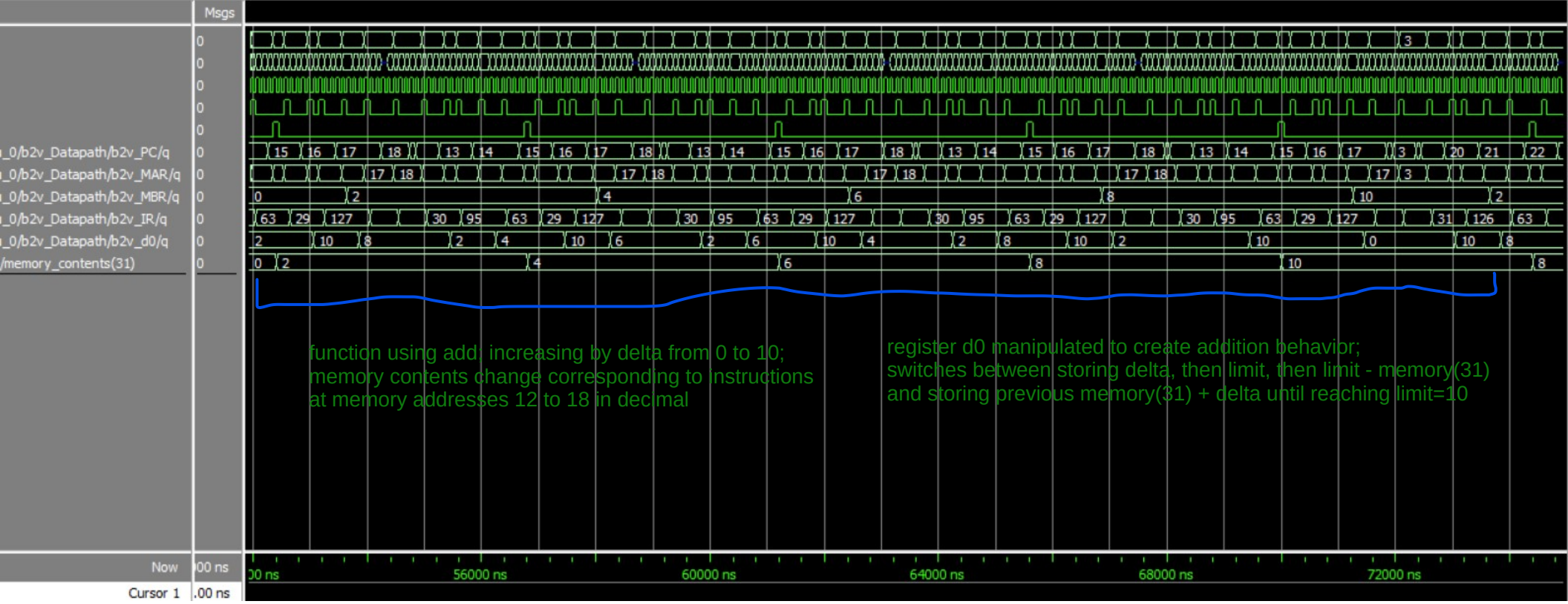
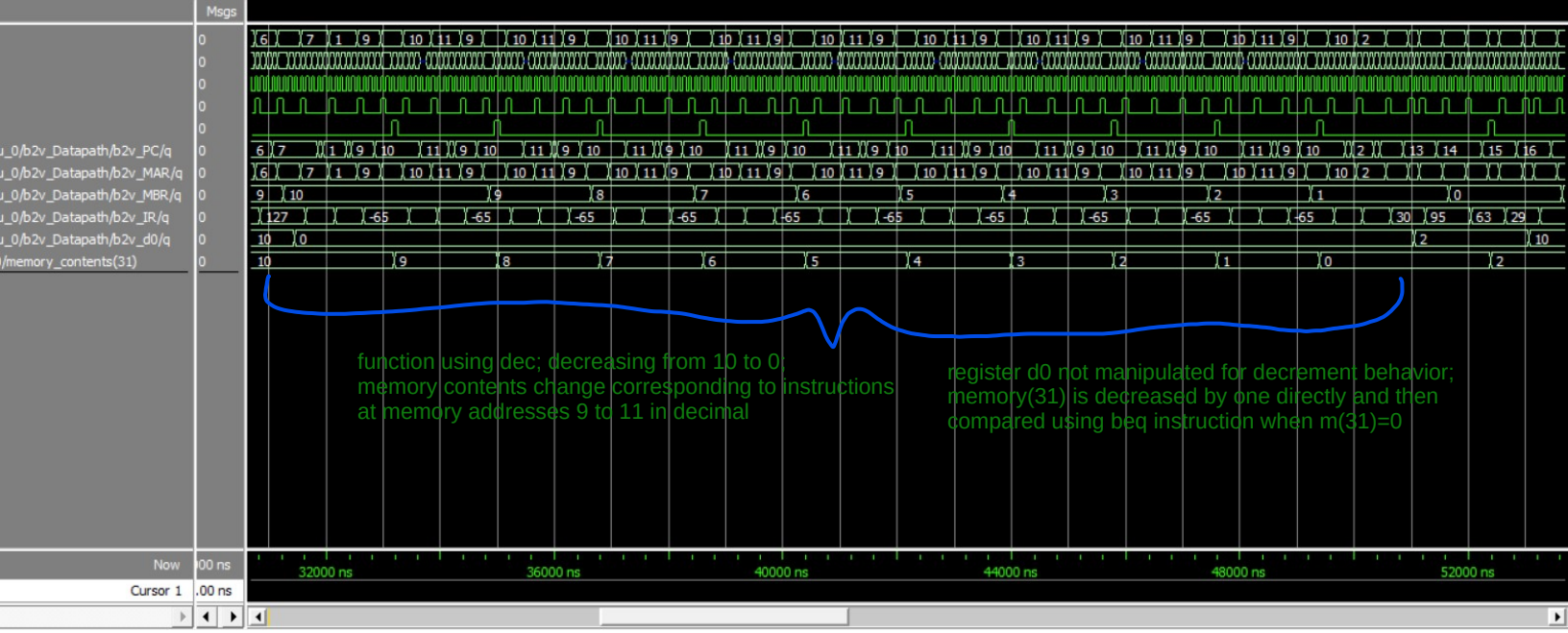
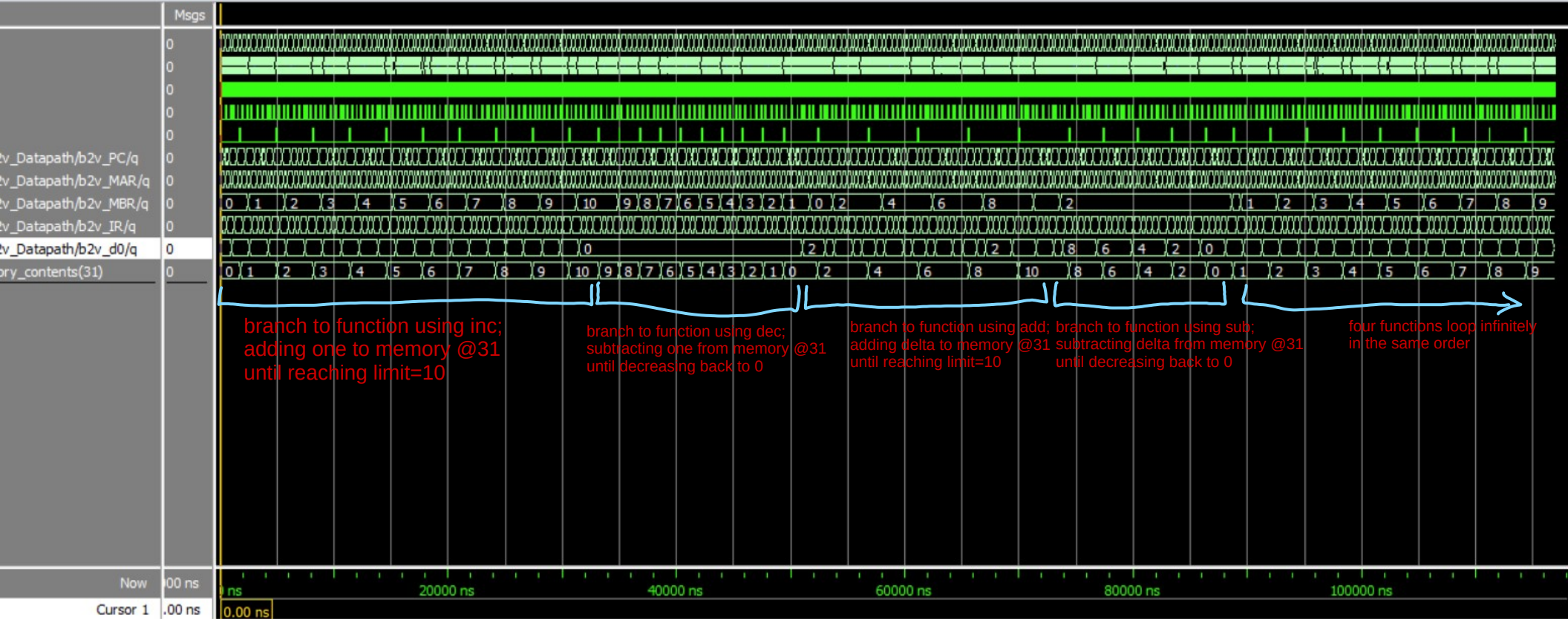
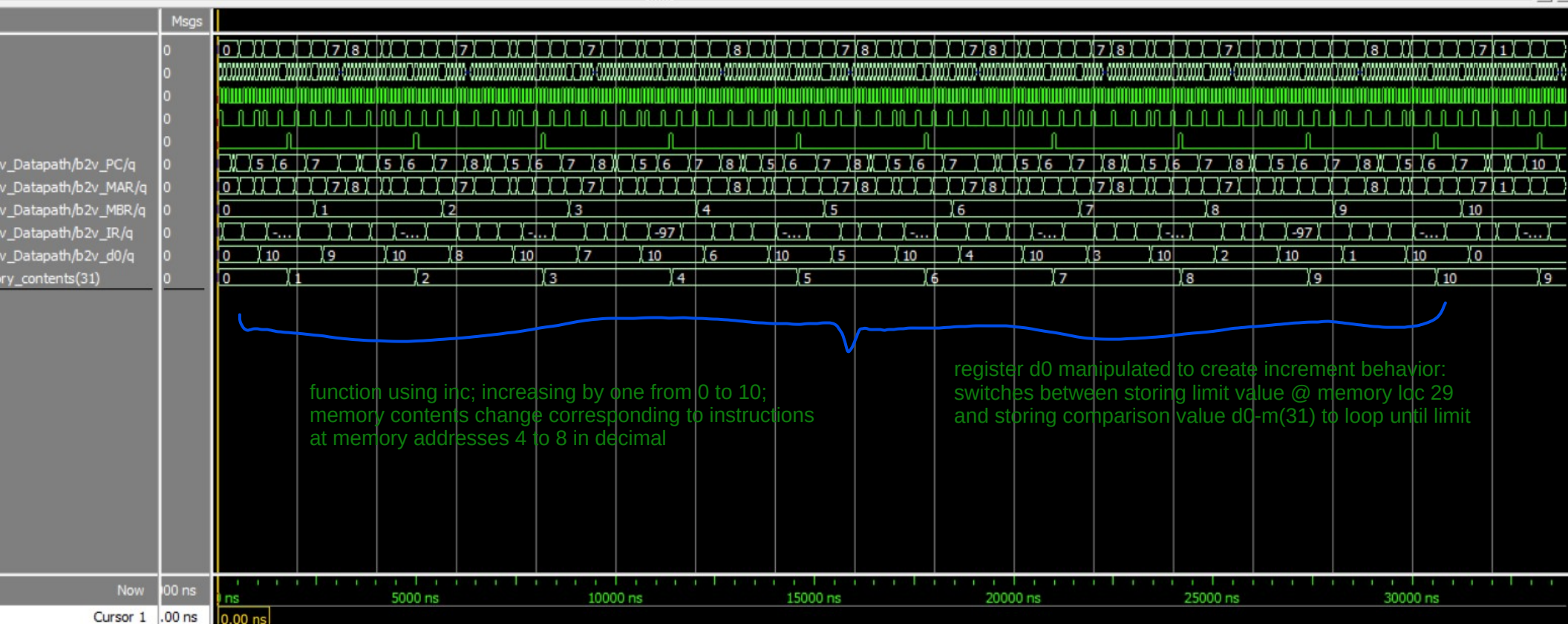
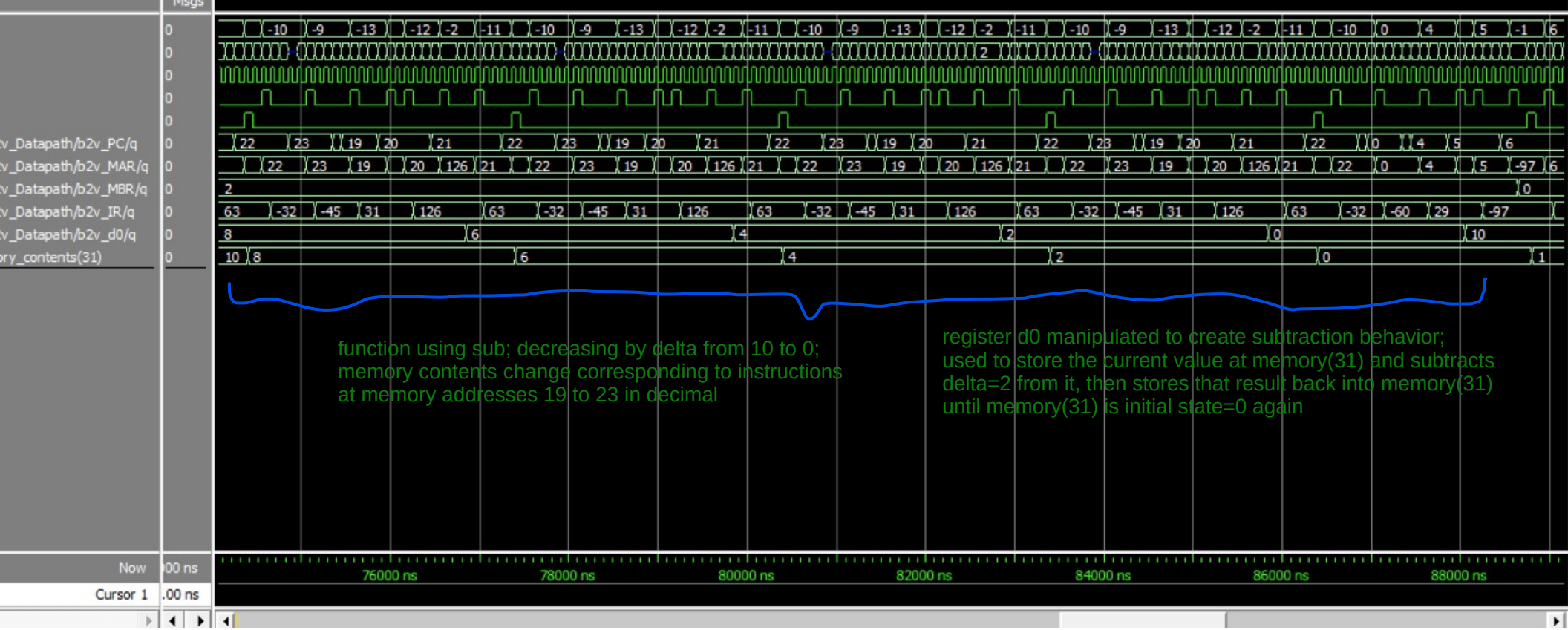

Msgs
_0/b2v_Datapath/b2v_PC/q
_0/b2v_Datapath/b2v_MAR/q
_0/b2v_Datapath/b2v_MBR/q
_0/b2v_Datapath/b2v_IR/q
_0/b2v_Datapath/b2v_d0/q
/memory_contents(31)
Now
Cursor 1
56000 ns
60000 ns
64000 ns
68000 ns
72000 ns
function using add; increasing by delta from 0 to 10;
memory contents change corresponding to instructions
at memory addresses 12 to 18 in decimal
register d0 manipulated to create addition behavior;
switches between storing delta, then limit, then limit - memory(31)
and storing previous memory(31) + delta until reaching limit=10


Msgs
u_0/b2v_Datapath/b2v_PC/q
u_0/b2v_Datapath/b2v_MAR/q
u_0/b2v_Datapath/b2v_MBR/q
u_0/b2v_Datapath/b2v_IR/q
u_0/b2v_Datapath/b2v_d0/q
/memory_contents(31)
Now
Cursor 1
32000 ns
36000 ns
40000 ns
44000 ns
48000 ns
52000 ns
function using dec; decreasing from 10 to 0;
memory contents change corresponding to instructions
at memory addresses 9 to 11 in decimal
register d0 not manipulated for decrement behavior;
memory(31) is decreased by one directly and then
compared using beq instruction when m(31)=0


Msgs
2v_Datapath/b2v_PC/q
2v_Datapath/b2v_MAR/q
2v_Datapath/b2v_MBR/q
2v_Datapath/b2v_IR/q
2v_Datapath/b2v_d0/q
ory_contents(31)
Now
Cursor 1
20000 ns
40000 ns
60000 ns
80000 ns
100000 ns
0.00 ns
branch to function using inc; adding one to memory @31 until reaching limit=10
branch to function using dec; subtracting one from memory @31 until decreasing back to 0
branch to function using add; adding delta to memory @31 until reaching limit=10
branch to function using sub; subtracting delta from memory @31 until decreasing back to 0
four functions loop infinitely in the same order


Msgs
v_Datapath/b2v_PC/q
v_Datapath/b2v_MAR/q
v_Datapath/b2v_MBR/q
v_Datapath/b2v_IR/q
v_Datapath/b2v_d0/q
ry_contents(31)
Now
Cursor 1
5000 ns
10000 ns
15000 ns
20000 ns
25000 ns
30000 ns
0.00 ns
function using inc; increasing by one from 0 to 10;
memory contents change corresponding to instructions
at memory addresses 4 to 8 in decimal
register d0 manipulated to create increment behavior:
switches between storing limit value @ memory loc 29
and storing comparison value d0-m(31) to loop until limit



function using sub; decreasing by delta from 10 to 0;
memory contents change corresponding to instructions
at memory addresses 19 to 23 in decimal

register d0 manipulated to create subtraction behavior;
used to store the current value at memory(31) and subtracts
delta=2 from it, then stores that result back into memory(31)
until memory(31) is initial state=0 again